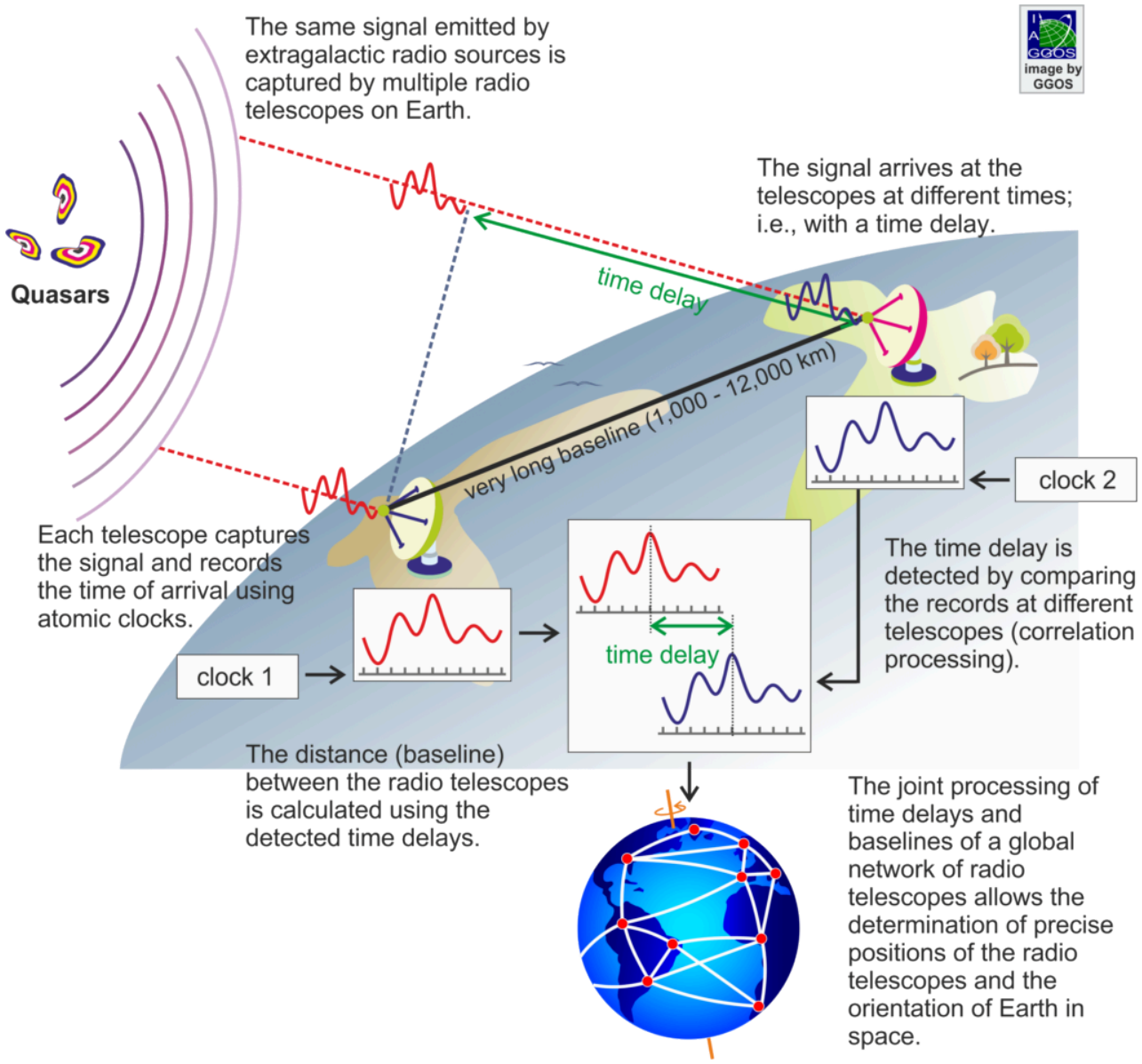
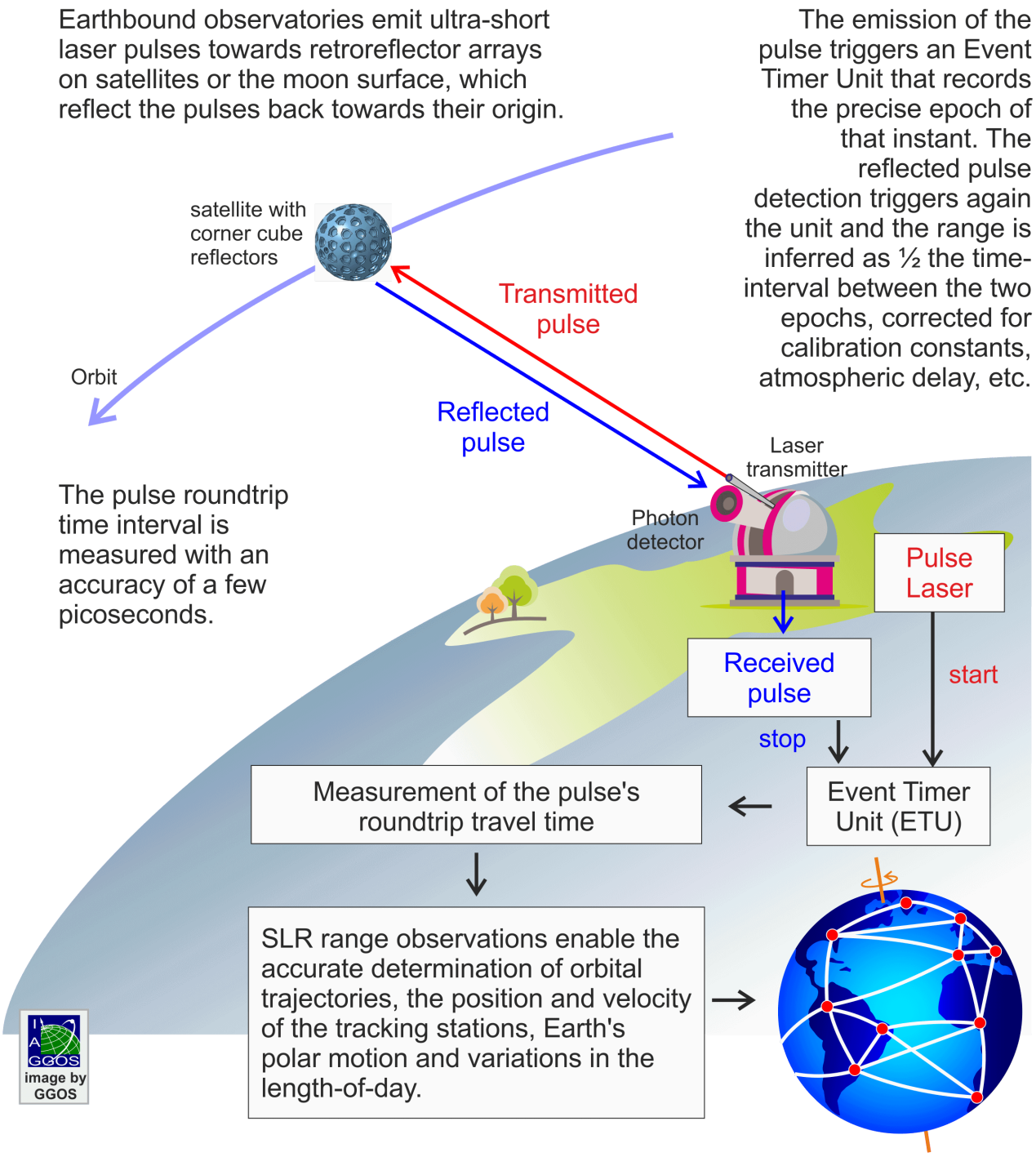
L3: International Terrestrial Reference System

- Geodetic reference systems required as an absolute reference
- Geodetic reference systems as a metrological basis
- Terrestrial reference system
  - Cartesian coordinates
  - Earth-fixed
  - Co-rotating with Earth
  - Defining:
    - Origin: Geocenter
    - Scale: SI meter
    - Orientation
- ITRF2020 latest ITRF
  - **Combination** of DORIS, GNSS, SLR, VLBI measurements
- **VLBI**



- Importance:
  - For ITRF scale
  - ICRF - extragalactic radio sources
  - 5 EOPs

- **SLR**

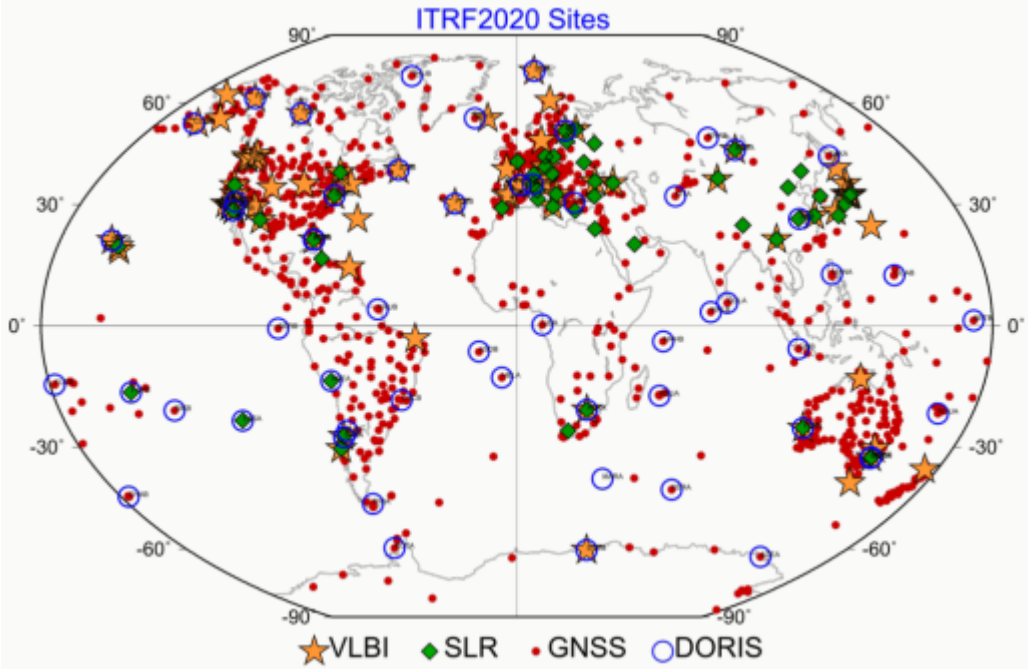


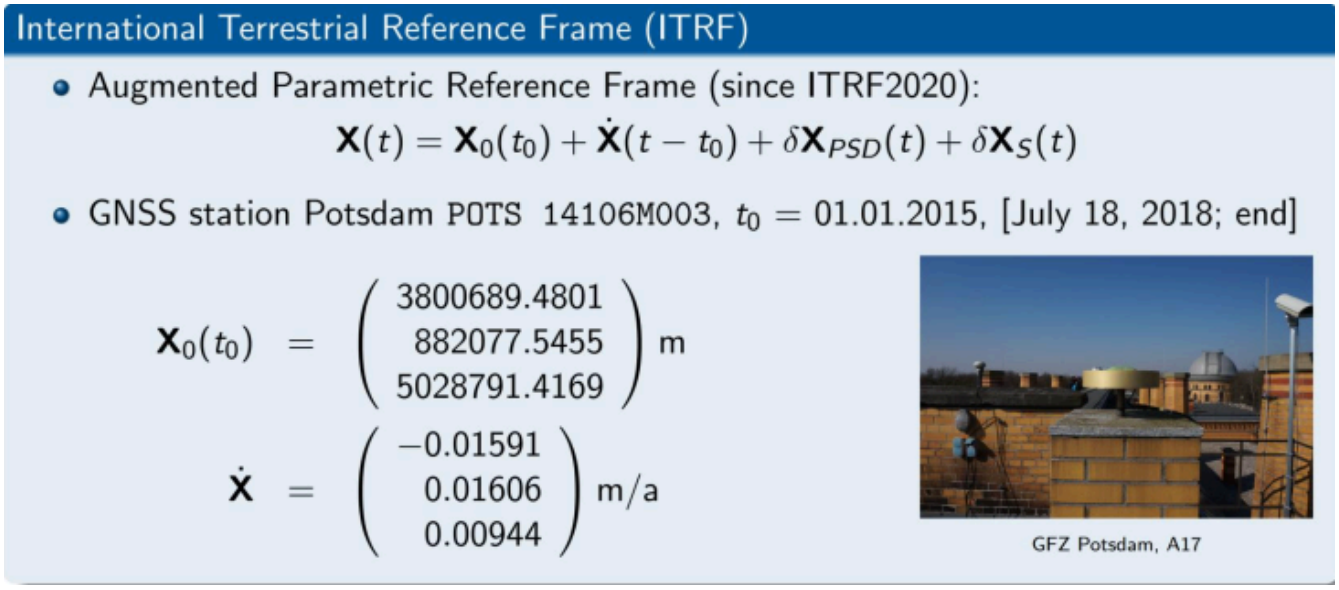
-
  - Origin of TRF
  - With VLBI -> Scale
- **GNSS**
  - Applications for high-precision Earth observation
- **DORIS**
  - Precise orbit determiantion of Low Earth Orbiting (LEO) satellites
  - Small and low cost
  - Best station network w.r.t. global and homogeneous distribution

Current Combination Strategy

- Combination of 4 space geodetic techniques via local measurements on the ground:

- **Local ties**
  - Co-located sites to connect different techniques

- ITRF Network
- 

- 

Realisations

- 3 combination centers of IERS
  - Deutsches Geodätisches Forschungsinstitut (DGFI-TUM)
    - DTRF2020
  - Institut National de l'Information Géographique et Forestière (IGN)
  - ITRF2020
- Jet Propulsion Laboratory (JPL)
  - JTRF2020

- Responsible to provide TRF products by combining ITRF inputs from the Technique Centres and others
- Regional Densifications for each Continent

State-Of-The-Art

- Geodetic reference frame as metrological basis for Earth system monitoring
  - Requirements 1mm - SotA: 5mm origin, 1mm scale
  - Long-term stability - SotA: 1mm/dec - 5mm/dec
- Goals by Global Geodetic Observing System
- SotA: ITRF2020

Summary

- Reference frame as the metrological basis for Earth system monitoring
- Combination of GNSS, SLR, VLBI, and DORIS to determine accurate and stable global terrestrial reference frames
- Current realizations: ITRF2020, DTRF2020, JTRF2020
- Current accuracy does not fulfill required accuracy → Limitations
- Investigation of technique-specific systematics and combination strategy essential (current research)